

Abdul Rehman Amer Baig

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Summary

Backend engineer with expertise in scalable APIs, cloud architecture (AWS), and payment systems (Stripe, Mollie). Experienced architecting AI agent pipelines with MCP tool-calling for healthcare automation, building freemium platforms with user-tier restrictions, and leading technical initiatives that improve team productivity. Competitive programmer with ICPC Silver Medal.

Experience

Software Engineer I, CloudEva

Mar 2025 – Present

Healthcare MVP

- Reduced prescribing medical template generation from **2 days to 2 hours** by building an **AI layer** using **FastMCP**, **Pydantic AI**, and **Claude**, eliminating manual engineering overhead through automated template creation. *Recognized with Certificate of Appreciation and launch celebration.*
- Integrated **Stripe** for feature subscriptions and **Mollie** for patient payments with vendor account connectivity and 10% stakeholder commission splitting.
- Resolved backend bottlenecks blocking the asynchronous event loop; redesigned authentication with **refresh tokens** and optimized database queries with proper indexing.
- Led backend development of **scalable APIs** and **system architecture** managing practitioner workflows, email flows via **AWS SES**, and secure document storage through **S3** with **AWS KMS** encryption-at-rest for data protection and compliance.
- Authored a document on indexing and backend best practices at management's request, sharing optimizations to enhance scalability across the company.

Athlete Jobs & Engagement Platform

- Launched freemium version with **user-tier restrictions**, **feature gating**, and role-based access controls.
- Integrated **HubSpot CRM** for lead tracking and **Twilio SMS** to automate personalized job alerts with quota-aware throttling and intelligent time gaps.
- Architected **AWS EventBridge** with **Connections** and **API Destinations** for timezone-aware news scheduling and publishing.
- Deployed **Redis** caching layer reducing job-service latency by 60% with **LRU** eviction policies.
- Decoupled authentication into a **Firestore-backed microservice**, creating a unified entry point across repositories and reducing code duplication.

Technical Leadership & Cross-Functional Collaboration

Demonstrated strong cross-functional leadership collaborating with designers, PMs, and engineering teams. Led full product modules from planning through deployment, presented weekly development updates to clients, and proactively resolved technical concerns while managing requirement clarification.

Academic Awards & Achievements

Silver Medalist, ICPC Onsite Regional Contest	2025
1st Runner Up, CODEFEST 3.0 Competitive Programming @ PUCIT	2023
1st Runner Up, Visio Spark Competitive Programming @ COMSATS	2024

Education

PUCIT (Punjab University College of Information and Technology)

2021 – 2025

BS in Data Science

- GPA: 3.68/4.0
- Coursework:** Object Oriented Programming, Data Structures & Algorithms, Web Development, Machine Learning
- Teaching Assistance:** Linear Algebra, Programming Fundamentals, Object Oriented Programming

Technical Skills

Core Technologies: Python, FastAPI, PostgreSQL, AWS, Stripe, Mollie, FastMCP, Pydantic AI, Git, GitHub

Familiar Technologies: Django, HTML, CSS, Bootstrap, NextJS